

Stockhunt Test Task — Walk-Forward + ML Results

Research universe: ~634 PIT S&P 500 constituents, 2008-01-02–2026-08-11. 7 of the 20 available indicators have rule-based trading strategies (see §2), spanning all four required families. Walk-forward: 5y in-sample / 1y out-of-sample / 1y rolling step, 14 windows, stitched OOS = 2013-01-01–2026-08-11 (final window is a partial ~7.5-month stub). Cost: 5 bps/side everywhere except the Phase 6 cost sweep. All numbers below are reproduced by `python run_all.py`.

1. Summary

- **Best strategy overall:** `combo_sma_rsi_or` (SMA 50/200 long-only OR RSI 21-day long-only), stitched OOS **Sharpe 0.89**, 95% bootstrap CI **[0.34, 1.48]**, net return 552%, max drawdown -40%, profitable on 89% of the 607 tickers it ever traded. Survives Benjamini-Hochberg correction across all 87 configurations evaluated in this project.
- **Wiring up volatility and volume indicators paid off, not just a checkbox.** Bollinger mean-reversion (volatility family, zero rule-based coverage before this pass) actually *won the in-sample sweep outright* (Sharpe 0.76 vs. SMA's 0.74), and both it (0.72 OOS) and MFI mean-reversion (0.62 OOS, volume family) landed ahead of the original RSI baseline (0.59 OOS) walk-forward. That said, SMA still wins walk-forward OOS despite losing the in-sample sweep — a clean illustration of exactly why walk-forward validation is required rather than trusting an in-sample ranking (see §8).
- **Regime-filter, the previously-missing 4th combination type, works — for one of the two pairings tested.** Gating SMA with an ADX trending-regime filter improved it (0.79 vs. 0.77 ungated) — the classic rationale held. Gating RSI mean-reversion with a low-volatility-regime filter *hurt* it (0.50 vs. 0.59 ungated) — the classic "mean-reversion works better in calm markets" rationale did not hold here; RSI's edge in this dataset appears concentrated in sharp, higher-volatility reversals (2020, 2022), not quiet range-bound stretches, so filtering those episodes out removed edge rather than noise. Reported as a real, non-obvious negative result, not smoothed over.
- **The long-side-more-robust hypothesis survives walk-forward, sharply:** the best RSI long-short config scored Sharpe 0.49 in-sample; walk-forward OOS it collapsed to **-0.03** (95% CI [-0.53, 0.43], not distinguishable from zero). Every long-only line held up; the one long-short line tested did not.
- **ML does not beat the best rule-based Sharpe.** Best ML model (logistic regression) scored 0.71 OOS vs. the best rule-based combo's 0.89. Per the task brief, this is reported as a strong result for the rule-based approach, not hedged. LightGBM did *worse* than the logistic baseline (Sharpe -0.08).
- **Neither the best rule-based nor the best ML strategy beats simply holding the research universe equal-weight (Sharpe 0.99) or SPY (Sharpe 0.92) over this OOS period.** This is the single most important qualifier on every other result in this report — see §6 and §10.
- **Every long-only rule-based strategy — old and new alike — is 0.91–0.95 correlated with the equal-weight basket benchmark.** They are close to a filtered version of market beta, not an independent source of return. The regime-filtered RSI+low-vol combo is the one clear exception (0.65 correlation) — restricting trading to specific volatility regimes makes it meaningfully less beta-like, even though it's also lower-Sharpe. The two ML models remain the only return streams in the whole study with near-zero correlation to everything else.

2. Method

- **Data foundation** (~6-8 hours, built before the work in the rest of this report): PIT S&P 500 membership reconstruction, Yahoo price download and ticker mapping (renamed/delisted securities), corporate-action and OHLC-consistency validation, missing/corrupted price history investigation. Several of its decisions are directly load-bearing for the no-look-ahead and cost-realism claims made throughout this report: a fixed 2026-08-11 research cutoff so later Yahoo data can never leak in; NYSE-session-aware next-day return construction (`build_next_day_returns`), including correct handling of extended exchange closures (e.g. Hurricane Sandy, Oct 2012), not a naive calendar-day shift; known-corrupted tickers excluded by name (`EXCLUDED_PRICE_TICKERS`); extreme returns flagged and retained, never clipped; and a signal without a valid next-session return treated as untradable, never a fabricated fill. Validated with an initial 34-config in-sample sweep before any walk-forward/ML/combination work began.
- **Indicator coverage:** `indicators.py` computes 20 indicators spanning trend, momentum, volatility, and volume. Before this pass, only 3 had rule-based trading strategies (`sma_crossover`, `rsi_mean_reversion`, `donchian_breakout` — trend and momentum only). This pass adds 4 more, one genuinely representative of each remaining gap: `macd_crossover` (trend), `stochastic_crossover` (momentum), `bollinger_mean_reversion` (volatility — previously zero coverage), `mfi_mean_reversion` (volume — previously zero coverage). **7 of 20 now have rules, not all 20** — a prioritized, family-complete subset given time constraints, stated honestly rather than presented as full coverage. The remaining 13 indicators are still used as ML features (§5).
- Phase 1 shortlist (10 configs: 6 curated + 1 programmatically-selected best-in-sample-long-only pick per new family) was screened by **in-sample Sharpe from the 66-config sweep**, not from inside the walk-forward loop itself. Stated as a deliberate, time-constrained compromise.

- Walk-forward re-optimizes over each strategy family's *full* parameter grid every window, so shortlist entries sharing a (family, mode) collapse to one walk-forward line — 10 shortlist entries produce 8 distinct lines, not 10.
- Combinations (Phase 3): 3 fixed-parameter pairs (SMA+RSI, SMA+Donchian, RSI+Donchian) × {AND, OR, weighted-vote}, plus 2 regime-filter pairs (SMA gated by an ADX trending-regime signal; RSI mean-reversion gated by a low-volatility-regime signal) — the 4th combination type named in the task brief, previously missing. `combine_signals(method="regime_filter")` in `src/walkforward/combo.py`.
- ML (Phase 4, unchanged by this pass — it already used all 20 indicators' worth of feature columns regardless of which had rule-based strategies): 30 of 35 available indicator columns, label = `sign(next-day return)`, purged (1-day) + embargoed (5-day) walk-forward CV aligned to the identical windows used everywhere else. No `train_test_split`/kFold anywhere. Trading rule: `P(up)≥0.70` long, `P(up)≤0.30` short, else flat.

3. Top strategies (rule-based, single)

line_id	Family	Sharpe	Net return	Max DD	% months profitable
<code>sma_crossover__long_only</code>	trend	0.774	355%	-37.7%	67.1%
<code>bollinger_mean_reversion__long_only</code>	volatility	0.724	445%	-43.0%	63.4%
<code>mfi_mean_reversion__long_only</code>	volume	0.621	321%	-46.1%	61.0%
<code>rsi_mean_reversion__long_only</code>	momentum	0.587	322%	-43.6%	60.4%
<code>macd_crossover__long_only</code>	trend	0.430	117%	-36.0%	61.6%
<code>stochastic_crossover__long_only</code>	momentum	0.028	-13%	-48.0%	56.7%
<code>rsi_mean_reversion__long_short</code>	momentum	-0.030	-11%	-33.3%	46.3%
<code>donchian_breakout__long_only</code>	trend	-0.855	-89%	-91.1%	46.3%

Every family now has a positive-Sharpe long-only representative except trend's second entry (MACD, still positive but weak) and momentum's second entry (Stochastic, essentially flat). Donchian remains the deliberately weak comparator and stayed weak out-of-sample. Stochastic crossover's best in-sample config was already negative (-0.13, see `results/shortlist.csv`) — the version shown here is its best walk-forward OOS run, still barely above zero. Not every indicator produces a working rule; reported as found, not cherry-picked.

4. Best combinations

line_id	Method	Sharpe	Net return	Max DD
<code>combo__sma_rsi_or</code>	OR	0.890	552%	-40.1%
<code>combo__sma_rsi_weighted_vote</code>	weighted (0.6 SMA / 0.4 RSI)	0.854	441%	-37.7%
<code>combo__sma_donchian_weighted_vote</code>	weighted (0.6 SMA / 0.4 Donchian)	0.854	441%	-37.7%
<code>combo__sma_donchian_or</code>	OR	0.818	401%	-37.7%
<code>combo__sma_adxtrend_regime_filter</code>	regime-filter (ADX-trend gate)	0.787	372%	-37.6%
<code>combo__rsi_donchian_weighted_vote</code>	weighted (0.6 RSI / 0.4 Donchian)	0.734	490%	-43.6%
<code>combo__sma_rsi_and</code>	AND	0.526	202%	-41.1%
<code>combo__rsi_lowvol_regime_filter</code>	regime-filter (low-vol gate)	0.503	216%	-48.4%
<code>combo__rsi_donchian_or</code>	OR	0.417	131%	-42.5%
<code>combo__rsi_donchian_and</code>	AND	-0.178	-5%	-7.0%
<code>combo__sma_donchian_and</code>	AND	-1.101	-93%	-94.4%

OR and weighted-vote both improved on their best individual leg in every pair; AND was worse than the better leg in every pair — requiring both signals to agree filtered out too many genuinely profitable trades. The ADX-trend regime filter on SMA (0.787) *beat* plain SMA (0.774) — turnover fell (fewer, better-timed trades) without giving up much upside, a genuine diversification-via-selectivity result. The low-vol regime filter on RSI (0.503) *underperformed* plain RSI (0.587) — see §1 for why. `combo__sma_donchian_or` matches `...weighted_vote` almost exactly because Donchian is so rarely active that OR and a 0.6/0.4 vote resolve to nearly the same trades.

5. Rule-based vs. ML head-to-head

line_id	Model	Sharpe	Net return	Max DD	Tickers traded	% profitable
<code>ml_logistic_regression</code>	Logistic Regression	0.711	870%	-45.2%	30	53.3%
<code>ml_lightgbm</code>	LightGBM	-0.079	-53%	-73.7%	—	—
<i>(best rule-based, for reference)</i> <code>combo__sma_rsi_or</code>	—	0.890	552%	-40.1%	607	89.1%

The best rule-based Sharpe (0.89) beats the best ML Sharpe (0.71) — reported as a strong result for the rule-based approach, not hedged, per the task brief.

LightGBM underperformed its own simpler logistic baseline. Permutation feature importance shows why: logistic regression's top features are `price_sma_dist_50` and `rsi_14` — directly the same signals that won as standalone rules (SMA and RSI both being the walk-forward top performers), i.e. convergent evidence. LightGBM's top features are `cmf_20`, `roc_20`, `volume_roc_20` — volume-based indicators that did not independently win as rules anywhere in this study, consistent with it having found relationships that generalized poorly. Notably, neither model's top features are `bb_zscore_20_2.0` or `mfi_14`, despite Bollinger and MFI being the two next-best standalone rules found this pass — the ML models and the rule search surfaced *different* signal, not fully overlapping evidence. See `results/charts/ml_feature_importance.png`.

The logistic model's 0.71 Sharpe is fragile in a way its headline number doesn't show: only 53% of months were profitable, and it only ever took a position in 30 of the 634 tickers in the universe (avg. gross exposure 7.4%) — a threshold-triggered, low-frequency, concentrated strategy. Its equity curve (`results/charts/equity_curves.png`) is visibly a staircase of long flat stretches and a few large jumps (notably 2024), not a smooth compounding line.

Feature-representation ablation (`src/pipeline/run_ml_pit_percentile_ablation.py`, `results/ml_pit_percentile_ablation.csv`, not part of the primary result above): replacing raw indicator values with same-date cross-sectional percentile ranks (`build_ml_dataset(..., use_percentile_rank=True)` in `src/ml/dataset.py`) is a principled normalization — it makes a feature comparable across tickers regardless of absolute scale or a market-wide regime shift — but it broke the fixed 0.70/0.30 trading threshold rather than improving on it. Percentile ranks are uniform on [0,1] by construction, with no fat tails, so logistic regression's predicted P(up) compressed to **[0.453, 0.558]** — it never once crossed 0.70 across 89,170 first-fold predictions, i.e. the model made zero trades. LightGBM widened slightly to [0.264, 0.656] but still only fired 4 times out of 89,170. The threshold was calibrated implicitly against the raw feature distribution; changing the feature representation without re-calibrating the threshold together with it silently disables the trading rule. Kept as a documented ablation rather than adopted as the primary result — recalibrating the threshold under the new distribution would mean abandoning the literal 0.70/0.30 framing, which was a deliberate choice (§2), not something to trade away for a normalization technique that, on this evidence, doesn't actually improve the outcome here.

6. Benchmarks

	Sharpe	Net return	Max DD
Equal-weight universe basket	0.992	829%	-40.3%
SPY buy-and-hold	0.916	566%	-33.7%
(best of this study, for reference) <code>combo__sma_rsi__or</code>	0.890	552%	-40.1%

Both benchmarks beat every single strategy tested in this project — old and new indicators alike — over the same stitched OOS window. This is the headline caveat on every result above — see §10.

7. Cost sensitivity

Best rule-based and best ML finalist re-run at 0/5/10/15/20/25 bps (0 bps used only here, per the hard constraint) — unchanged from before this pass since neither finalist changed:

Cost (bps)	<code>combo__sma_rsi__or</code> Sharpe	<code>ml_logistic_regression</code> Sharpe
0	0.915	0.726
5 (baseline)	0.890	0.711
10	0.865	0.696
15	0.840	0.682
20	0.815	0.667
25	0.790	0.652

Neither edge dies anywhere in this range. Chart: `results/charts/cost_sensitivity.png`.

8. Walk-forward efficiency and ranking stability

line	avg(OOS Sharpe / IS Sharpe)	avg rank corr (IS vs OOS)	% windows IS winner = OOS best
<code>sma_crossover__long_only</code>	1.51	0.11	21%
<code>bollinger_mean_reversion__long_only</code>	1.27	0.29	29%
<code>mfi_mean_reversion__long_only</code>	1.20	0.03	36%
<code>rsi_mean_reversion__long_only</code>	1.05	0.23	7%
<code>macd_crossover__long_only</code>	2.84	0.39	21%

line	avg(OOS Sharpe / IS Sharpe)	avg rank corr (IS vs OOS)	% windows IS winner = OOS best
stochastic_crossover__long_only	-7.92	0.24	36%
rsi_mean_reversion__long_short	0.10	0.07	14%
donchian_breakout__long_only	n/a (IS Sharpe never positive)	0.11	0%

Bollinger beating SMA in-sample but losing to it out-of-sample (§1) is the clearest illustration in this dataset of why the task brief requires walk-forward rather than an in-sample ranking — an in-sample-only report would have shipped Bollinger as the headline strategy. Stochastic's -7.92 "efficiency" is an artifact of a near-zero in-sample Sharpe denominator, not a meaningful ratio — included for completeness, not implying an 8x blowup is real. Ranking stability is weak everywhere (rank correlations 0.03–0.39) — the parameter grids are small and per-window differences between candidates are mostly noise.

9. Robustness

Block-bootstrap Sharpe CIs and multiple-testing correction (1000 resamples, 20-day blocks; Benjamini-Hochberg at $\alpha=0.05$ across the 21 walk-forward/OOS finalists — out of **87 configurations tested in total**: 66 in-sample sweep + 21 OOS finalists; the sweep is disclosed for context, not re-tested since it has no OOS return series to bootstrap):

13 of 21 finalists remain significant after correction, including the top result (combo__sma_rsi__or, $p=0$, 95% CI [0.34, 1.48]), both new standalone strategies (Bollinger $p=0.014$, MFI $p=0.015$), the ADX-regime-filter combo ($p=0.003$), the low-vol-regime-filter combo despite its lower raw Sharpe ($p=0.037$), and the ML logistic model ($p=0.003$). **Not significant**: MACD ($p=0.069$, just misses), Stochastic, both long-short lines, ml_lightgbm, and both Donchian-involving AND combos. Full table: results/robustness_bootstrap.csv.

Parameter-stability: the original sparse 6-point SMA sweep grid understated this — its nearest tested neighbor to the 50/200 winner was 30/100, a real gap (0.74 vs. 0.57) that couldn't rule out a local spike. A dense 5×5 local grid was built around each of the 5 finalists with genuine positive Sharpe (SMA, Bollinger, MFI, RSI, MACD — Donchian, Stochastic, and RSI long-short excluded as non-winners), varying each strategy's two structural parameters while holding the rest fixed at their winning values, in-sample, same methodology as the original sweep. **All five show a real plateau, not a spike**: SMA ranges 0.62–0.78 (spread 0.16) across the whole 5×5 neighborhood, MACD 0.26–0.33 (spread 0.065, the tightest of the five), RSI 0.47–0.77 (spread 0.31), Bollinger 0.46–0.82 (spread 0.35), MFI 0.51–0.71 (spread 0.20) — every cell in every grid stays positive, and each original winner sits inside its plateau rather than at an isolated peak (Bollinger's grid actually found a nearby point 0.06 Sharpe *better* than the original shortlisted pick — window=17/entry=-0.5 vs. window=20/entry=-1.5 — evidence the original coarse sweep simply hadn't sampled that neighborhood, not that the result is fragile). Charts: results/charts/param_stability_{sma_crossover,macd_crossover,rsi_mean_reversion,bollinger_mean_reversion,mfi_mean_reversion}.png.

Breadth: computed for all 21 tested strategies (results/breadth_summary.csv), not just the top finalist. combo__sma_rsi__or leads at 541/607 (89.1%); the next four are also combos (83–84%). Breadth and Sharpe don't always agree: stochastic_crossover has high breadth (78.8%) despite ~zero Sharpe (many small winners, no edge after costs), and donchian_breakout/ml_lightgbm show the opposite pattern — donchian_breakout is profitable on 49.4% of names yet has the worst Sharpe and drawdown in the study (a few catastrophic losers overwhelm many modest winners); ml_lightgbm is profitable on 58.1% of the (few) names it traded yet still lost money overall. ml_logistic_regression remains the narrowest bet in the study: 16/30 (53.3%).

Survivorship bias, quantified (the PDF's explicit ask, not previously closed): re-running combo__sma_rsi__or's exact stitched OOS signal, unchanged, restricted to only the 474 tickers that are S&P 500 constituents *today* (vs. the full 611-ticker PIT-reconstructed universe the signal actually traded) — same window, same cost — **survivorship bias overstates the Sharpe ratio by +0.084 (0.893 → 0.977) and net return by +108 percentage points (557% → 665%)**. Direction: using today's list instead of the correct PIT list makes a strategy look better than it is, as expected — a non-PIT-aware backtest on this data would have overstated the edge by roughly 9% of the Sharpe itself. results/survivorship_bias_quantification.csv.

10. Where results are weakest

- **No strategy in this study beat its own benchmark**, old or new. The equal-weight basket (0.99 Sharpe) and SPY (0.92) both beat the best combo (0.89) and the best ML model (0.71).
- **The rule-based edge is mostly beta, not alpha, and this held even after adding 4 new indicator families**. Every long-only rule/combo tested — SMA, RSI, MACD, Bollinger, MFI, Stochastic, and their combinations — is 0.91–0.95 correlated with the equal-weight basket (§1/§9, correlation_matrix.csv). The one exception, the low-vol-regime-filtered RSI combo (0.65 correlation), is also one of the lower-Sharpe results — the diversification and the underperformance appear to be the same phenomenon (trading a narrower slice of market conditions).
- **7 of 20 indicators have rules, not all 20**. Trend and momentum each have a second, weaker representative (MACD, Stochastic); volatility and volume each have exactly one (Bollinger, MFI). The remaining 13 are ML features only. Stated explicitly rather than left implicit.

- **In-sample-screened shortlist** (§2): the shortlist that entered walk-forward was chosen by in-sample Sharpe. Bollinger's in-sample win over SMA (§8) shows this ranking is not reliable — a strategy screened only inside the walk-forward loop from the start might have produced a different top set.
- **ML logistic result is concentrated**: 30 tickers, 53% of months profitable, staircase equity curve.
- **Regime-filter's second pairing (RSI + low-volatility gate) underperformed the ungated strategy** — included and explained (§1, §4), not dropped for looking bad. Only 2 regime-filter pairings were tested; a broader sweep of gate thresholds or gate/base-strategy pairings was out of scope for this pass.
- **Turnover-at-roll-date artifact**: `run_backtest` computes turnover from a weight-matrix pivot of only the rows it's given, so day 1 of every stitched OOS segment looks like a from-flat entry rather than a continuation of the prior window's book — a small, documented mechanical overstatement of turnover/cost at each of the 13 roll dates. Not a look-ahead leak.
- **Weak ranking stability** (§8): the walk-forward re-optimization step is not clearly finding a durable signal above noise at this grid resolution.
- The two stateful mean-reversion state machines with the slowest per-call runtime (`_rsi_target_state_long_only/_long_short`, reused by MFI; the Bollinger equivalent) are plain per-ticker Python loops, not vectorized — functionally correct (see `tests/test_no_lookahead.py` and the smoke tests) but a real performance gap, not just a style choice: the dense parameter-stability grids for Bollinger and MFI took most of this pass's total runtime for exactly this reason.
- **Cross-sectional PIT percentile normalization was attempted, not skipped** (§5 ablation) — tried, diagnosed as breaking the fixed confidence threshold, not adopted. Still genuinely not attempted: SHAP feature importance (permutation importance used instead); long-short combos; regime-filter pairings beyond the 2 tested; wiring the remaining 13 indicators as rules; PCA/SHAP/Boruta-driven feature reduction ahead of model training (5 redundant columns were dropped by manual inspection instead, see `src/ml/dataset.py`); ML hyperparameter tuning (both models use fixed, untuned defaults); a re-calibrated threshold that would make the percentile-normalized features usable (e.g. per-window quantile-based cutoffs instead of a fixed value).

Appendix: reproducing these numbers

`python run_all.py` from a clean `results/` directory reproduces every number in this report. See `README.md` for phase-by-phase output locations and the column glossary.